PSI Data Organization Document

Advanced Combustion via Microgravity Experiments (ACME) Structure and Response of Spherical Diffusion Flames (s-Flame)

To gain an initial understanding of this experiment, refer to the documents “ACME non-imaging data file description.pdf” and “s-Flame_final_report_20210729.pdf” located in “Final Report” of the “Resulting Products” section on the “Pubs & Docs” tab.

The s-Flame data is organized in PSI by diluent and fuel composition. Raw data, and video & image data are available on the “Scientific Data and Information Tab” of each of the eight “Experiment Data” records:

- 1 Nitrogen-diluted Hydrogen and Oxygen
- 2 Nitrogen-diluted Methane and Oxygen
- 3 Nitrogen-diluted Hydrogen-Methane mixtures and Oxygen
- 4 Nitrogen-diluted Ethylene and Oxygen
- 5 Helium-diluted Hydrogen and Oxygen
- 6 Helium-diluted Methane and Oxygen
- 7 Helium-diluted Hydrogen-Methane mixtures and Oxygen
- 8 Helium-diluted Ethylene and Oxygen

Within each “Experiment Data” record, “Raw Data”, “Images” and “Audio/Video Files” are available on the “Scientific Data and Information” tab. Test names are formatted YYDDDX# where YY identifies year, DDD identifies day of year (GMT), X identifies a letter and # identifies a number – where the different numbers can be for repeats but are typically variants, e.g., with different flow rate(s) or instrumentation setting(s), e.g. 19308F1. Record naming and listings are based on the detailed information in the test conditions table below (page 2).

Test specific raw data files are located on the “Raw Data and Information” tab of the “Raw & Analyzed Data” record. The first row of the .csv files provides information about the test configuration. The headers of the basic data in .csv files is explained in Section 10.1, beginning on page 126 of the “ACME-DOC-010 Rev D.pdf” located in the “Data Organization Document(s)” section of the “Scientific Data and Information” section on the “Data” tab of the ACME s-Flames “Investigations” record. Associated videos are located on the “General” tab and associated images are located on the “Image Data and Information” tab of the “Videos & Images” record.

There are various other supporting engineering documents, videos, reports and publications available on the other tabs of the “Investigations” record.

Table 1. Flames ignited for the s-Flame experiment

Test Day[1]	Year	Month	Date	Atmosphere[2]	Fuel bottle	Flames ignited[3]
First round of tests, with nitrogen-diluted fuel and oxygen						
19254	2019	9	11	~21% O_2 with N_2	50% C_2H_4 with N_2	13
19256			13			16
19304		10	31		45% CH_4 with N_2	9
19308		11	4	~30% O_2 with N_2		21
19352		12	18	~21% O_2, with N_2	100% H_2	10
20006	2020	1	6		44.4% H_2 with CH_4	17
20013			13		66.7% H_2 with CH_4	22
20024			24		100% H_2	16
20029			29		50% C_2H_4 with N_2	21
First round of tests, with helium-diluted fuel and oxygen						
20132	2020	5	11	~21% O2 with He	25% H2, 20% CH4, with He	7
20136			15		20% H2, 25% CH4, with He	11
20142			21	~30% O2 with He	30% H2, 15% CH4, with He	17
20155		6	3	~30% and ~21% O2 with He	45% H2 with He	9
20160			8		45% CH4 with He	12
20164			12	~30% O2 with He	20% C2H4 with He	11
20170			18	~30% and ~21% O2 with He	25% C2H4 with He	20
20176			24		30% C2H4 with He	23
20190		7	8		45% H2 with He	21

[1]Test Day includes the year as the first two digits followed by the 3-digit day of the year.
[2]Atmosphere is ~1 atm but pressure increases and O2% drops with vitiation.
[3]Flames ignited is the number of nominally sustained flames ignited on that day (neglecting the day change on ISS).